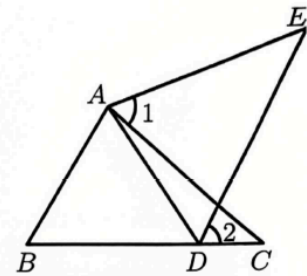


L3 E

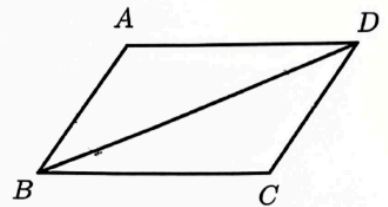
- 9 As shown in the figure below, $m\angle 1 = m\angle 2$, in order to prove $\triangle ABC \cong \triangle ADE$, we can add condition _____.



- A. $AB = AD, AC = AE$
C. $AB = DE, BC = AE$

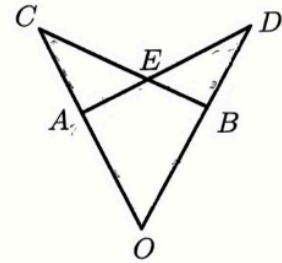
- B. $AB = AD, BC = DE$
D. $AC = AE, BC = DE$

- 10 As shown in the figure below, $AB = CD$, $\overline{AB} \parallel \overline{CD}$. Prove that $AD = BC$.



L4

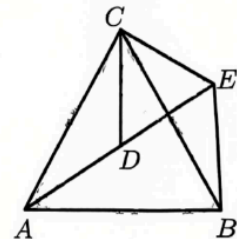
- 6 As shown in the diagram, $OA = OB$, $OC = OD$, $m\angle O = 60^\circ$, $m\angle C = 25^\circ$. The measure of $\angle BED$ is _____.



.. 70°
 C. 65°

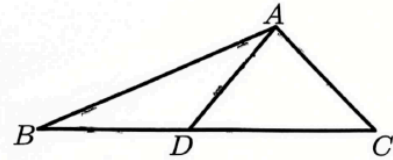
B. 85°
 D. None of the above

- 13 As shown in the figure below, $\triangle ACB$ and $\triangle DCE$ are equilateral triangles. D and E are on the same line. Connect $\overline{BE}$. $m\angle AEB$ is _____.

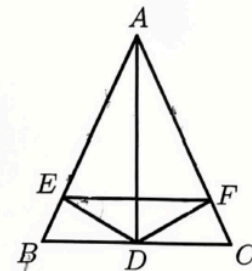


L5 E

- 2 As shown in the figure, in $\triangle ABC$, D is on $\overline{BC}$ and $AC = AD = DB$. If $m\angle BAC = 102^\circ$, then $m\angle ADC = \underline{\hspace{1cm}}^\circ$.



- 3 As shown in the figure below, in $\triangle ABC$, $AB = AC$, $\overline{AD}$ bisects $\angle BAC$, $\overline{DE} \perp \overline{AB}$ at E and $\overline{DF} \perp \overline{AC}$ at F . Among the four following conditions: (1) $m\angle DEF = m\angle DFE$; (2) $AE = AF$; (3) $\overline{AD}$ bisects $\angle EDF$; (4) $\overline{EF}$ perpendicular bisects $\overline{AD}$, how many of them are correct?



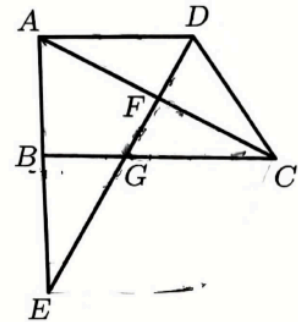
A. 1

B. 2

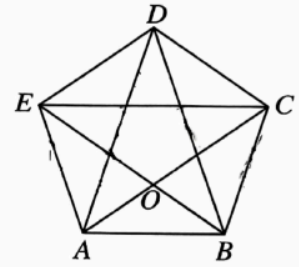
C. 3

D. 4

- 8 As shown in the figure, in quadrilateral $ABCD$, $\overline{AD} \parallel \overline{BC}$, $m\angle ABC = 90^\circ$, $\overline{DE} \perp \overline{AC}$ at point F , intersecting $\overline{BC}$ at point G and $\overline{AE}$ at E . $AE = AC$. Prove that $BG = FG$. (Hint: Connect $\overline{AG}$)



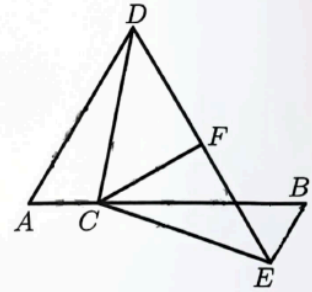
- 10 As shown in the figure below, $\overline{AE} \parallel \overline{BD}$, $\overline{BC} \parallel \overline{AD}$, $AE = BC$ and $AD = BD$, prove that:
- (1) $AC = BE$



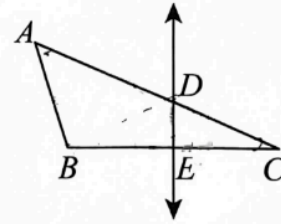
- (2) $\overline{EC} \parallel \overline{AB}$

L6

- 3 As shown in the figure below, point C is on segment AB , $\overline{AD} \parallel \overline{EB}$, $AC = BE$, $AD = BC$, $\overline{CF}$ bisects $\angle DCE$. Prove: $\overline{CF} \perp \overline{DE}$.

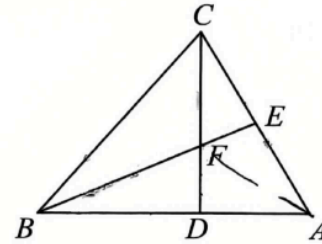


- 6 As shown in the figure below, in $\triangle ABC$, perpendicular bisector of $\overline{BC}$ intersects $\overline{AC}$ and $\overline{BC}$ at D and E , respectively. If $AB = CD$, $m\angle C = 20^\circ$, then $m\angle A = \underline{\hspace{1cm}}$.



- 8 As shown in the figure below, in $\triangle CBD$, $CD = BD$, $\overline{CD} \perp \overline{BD}$, $\overline{BE}$ bisects $\angle CBA$ and intersects $\overline{CD}$ at F . $\overline{CE} \perp \overline{BE}$ at E , $\overline{CE}$ intersects $\overline{BD}$ at A .

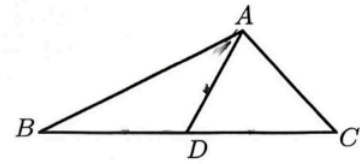
(1) Prove: $BF = AC$.



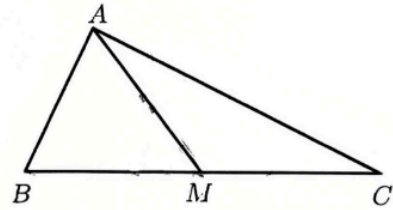
(2) $\overleftrightarrow{BE}$ is the perpendicular bisector of $\overline{AC}$.

L7

- 5 In $\triangle ABC$, $BD = CD$, $\angle DAC = \angle ABC + \angle ACB$. Given that $AD = 3$, find the length of $AB = \underline{\hspace{2cm}}$.

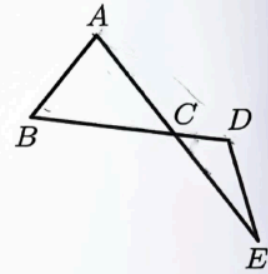


- 6 As shown in the figure, in $\triangle ABC$, $\overline{AM}$ is the median. Prove that $AM < \frac{1}{2}(AB + AC)$.

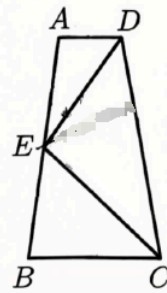


L7 E

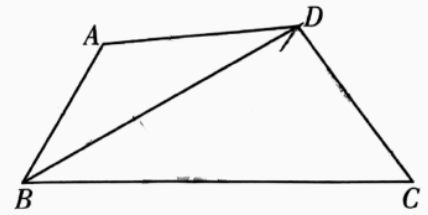
27 As shown in the figure below, $m\angle B + m\angle CDE = 180^\circ$, $AC = CE$. Prove: $AB = DE$.



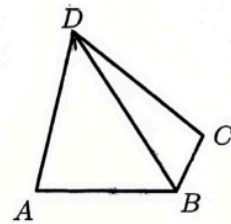
- 4 As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, E is the midpoint of $\overline{AB}$, $\overline{DE} \perp \overline{CE}$, $AD = 3$, and $CD = 9$. Then $BC = \underline{\hspace{2cm}}$.



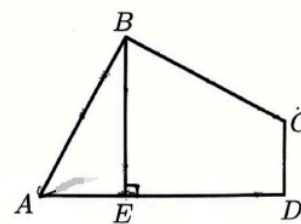
- 6 As shown in the figure, given that in quadrilateral $ABCD$, $BC > AB$, $AD = CD$, $\overline{BD}$ bisects $\angle ABC$, prove that $m\angle A + m\angle C = 180^\circ$.



- 7 As shown in the figure, in $ABCD$, $\overline{BD}$ bisects $\angle ABC$ and $m\angle A + m\angle C = 180^\circ$. If $AB > BC$, prove that $AD = DC$.



- 9 In the figure, in quadrilateral $ABCD$, $AB = BC$, $m\angle ABC = m\angle CDA = 90^\circ$, $\overline{BE} \perp \overline{AD}$ at E , and the area of quadrilateral $ABCD$ is 10. The length of $\overline{BE}$ is _____.

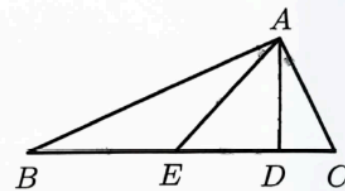


L8

- 8 In triangle ABC , $AB = AC$, $AC = 2$, and $m\angle BAC = 30^\circ$. Find the length of $\overline{BC}$.

L8 E

- 7 As shown in the figure, in $\triangle ABC$, $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at point D . E is the midpoint of $\overline{BC}$, and $AD = ED = 3$. The length of $\overline{BC}$ is _____.



A. $6\sqrt{2}$

B. $4\sqrt{3}$

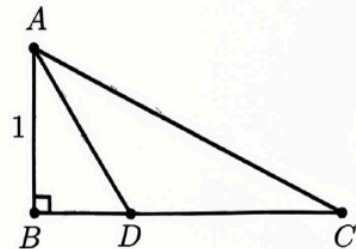
C. $3\sqrt{2}$

D. $3\sqrt{2} + 2\sqrt{3}$

- 2 Right triangle ABC has leg lengths $AB = 20$ and $BC = 21$. Including $\overline{AB}$ and $\overline{BC}$, how many line segments with integer length can be drawn from vertex B to a point on hypotenuse $\overline{AC}$? (2018 AMC 10A Problem, Question #16)

A. 5 B. 8 C. 12 D. 13 E. 15

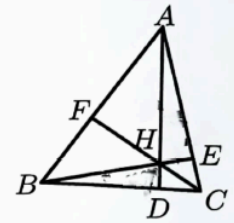
- 4 Triangle ABC has a right angle at B , $AB = 1$, and $BC = 2$. The bisector of $\angle BAC$ meets $\overline{BC}$ at D . What is BD ? (2009 AMC 10B Problem, Question #20)



A. $\frac{\sqrt{3}-1}{2}$ B. $\frac{\sqrt{5}-1}{2}$ C. $\frac{\sqrt{5}+1}{2}$ D. $\frac{\sqrt{6}+\sqrt{2}}{2}$ E. $2\sqrt{3}-1$

L9

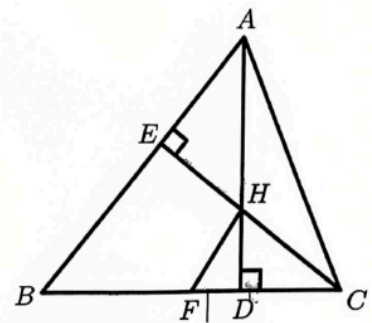
- 7 In the figure, H is the orthocenter (the intersection point of the three altitudes) of acute triangle ABC . If $AH = BC$, then the degree measure of angle BAC is _____.



L9 E

- 5 In the acute triangle ABC , let $\overline{AD}$ and $\overline{CE}$ be the altitudes, and let H be the orthocenter. Given that $AD = BC$ and F is the midpoint of $\overline{BC}$, prove that:

$$FH + DH = \frac{1}{2}BC.$$



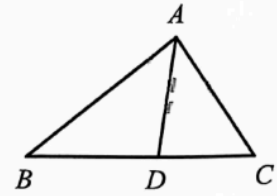
6 In the isosceles triangle $\triangle ABC$, $AB = AC = 6$, $BC = 6\sqrt{3}$, and I is the incenter of $\triangle ABC$.

(1) Find $m\angle BIC$.

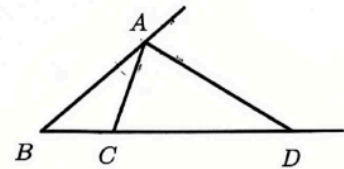
(2) Find AI .

8 Use parallel line to prove the following:

(1) As shown in the figure below, $\overline{AD}$ is the angle bisector of $\triangle ABC$, prove that $\frac{AB}{AC} = \frac{BD}{CD}$.

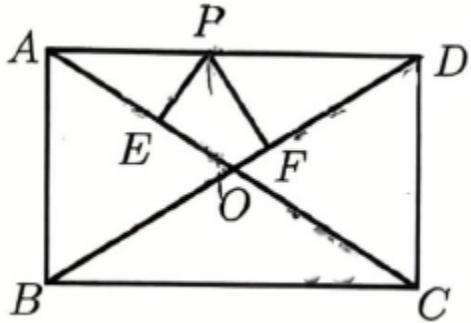


(2) In the figure, the external angle bisector of $\angle BAC$ intersects the extension of side BC at point D . Prove that $\frac{AB}{AC} = \frac{BD}{CD}$.



L10

- 9 As shown in the figure, in rectangle $ABCD$, $AB = 3$, $AD = 4$, and P is a moving point on $\overline{AD}$. Given that $\overline{PF} \perp \overline{BD}$ at point F and that $\overline{PE} \perp \overline{AC}$ at point E , then $PE + PF$ is _____.



- 6 As shown in the figure, the diagonals of the parallelogram $ABCD$ intersect at point O , and the line EF passes through the point O , intersecting with the extensions of $\overline{AB}$ and $\overline{CD}$ at the point E and F , respectively. Prove: The quadrilateral $AECF$ is a parallelogram.

